

lab5

CCS3400 G1

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Q1

- code

```
SELECT COUNT(DISTINCT department_name)
AS number_of_departments
FROM departments;
```

- result

	NUMBER_OF_DEPARTMENTS
1	8

Q2

- code

```
--Q2
SELECT department_id AS "Department ID",
SUM(salary) AS "Total Salary Paid"
FROM employees
GROUP BY department_id
ORDER BY department_id;
```

- Result

	Department ID	Total Salary Paid
1	10	4400
2	20	19000
3	50	17500
4	60	19200
5	80	30100
6	90	58000
7	110	20300
8	(null)	7000

Q3

- code

```
--Q3
SELECT
    ROUND(MAX(salary)) AS Maximum,
    ROUND(MIN(salary)) AS Minimum,
    ROUND(SUM(salary)) AS Sum,
    ROUND(AVG(salary)) AS Average
FROM employees;
```

- Result

	MAXIMUM	MINIMUM	SUM	AVERAGE
1	24000	2500	175500	8775

Q4

- code

```
--Q4
SELECT
    job_id,
    MIN(salary) AS Minimum,
    MAX(salary) AS Maximum,
    SUM(salary) AS Sum,
    ROUND(AVG(salary)) AS Average
FROM employees
GROUP BY job_id;
```

- Result

SQL All Rows Fetched: 12 in 0.022 seconds						
	JOB_ID	MINIMUM	MAXIMUM	SUM	AVERAGE	
1	AD_VP	17000	17000	34000	17000	
2	AC_MGR	12000	12000	12000	12000	
3	ST_CLERK	2500	3500	11700	2925	
4	AD_ASST	4400	4400	4400	4400	
5	IT_PROG	4200	9000	19200	6400	
6	SA_MAN	10500	10500	10500	10500	
7	AC_ACCOUNT	8300	8300	8300	8300	
8	ST_MAN	5800	5800	5800	5800	
9	AD_PRES	24000	24000	24000	24000	
10	MK_MAN	13000	13000	13000	13000	
11	SA_REP	7000	11000	26600	8867	
12	MK_REP	6000	6000	6000	6000	

Q5

- Code

```
--Q5
SELECT
    job_id,
    COUNT(*) AS number_of_people
FROM employees
GROUP BY job_id;
```

- result

	⚡ JOB_ID	⚡ NUMBER_OF_PEOPLE
1	AD_VP	2
2	AC_MGR	1
3	ST_CLERK	4
4	AD_ASST	1
5	IT_PROG	3
6	SA_MAN	1
7	AC_ACCOUNT	1
8	ST_MAN	1
9	AD_PRES	1
10	MK_MAN	1
11	SA_REP	3
12	MK_REP	1

Q6

- code

--Q6

```
SELECT  
    COUNT(DISTINCT manager_id) AS "Number of Managers"  
FROM employees;
```

- result

	Number of Managers
1	8

Q7

- code

```
--Q7  
SELECT  
    MAX(salary) - MIN(salary) AS DIFFERENCE  
FROM employees;
```

- result

	DIFFERENCE
1	21500

Q8

- code

--Q8

```
SELECT
    department_id,
    SUM(salary) AS total_salary
FROM employees
GROUP BY department_id
HAVING SUM(salary) > 25000;
```

- result

	DEPARTMENT_ID	TOTAL_SALARY
1	90	58000
2	80	30100

Q9

- code

```
--Q9
SELECT
    department_id,
    AVG(salary) AS "Average Salaries"
FROM employees
WHERE department_id IN (50, 60)
GROUP BY department_id;
```

- result

	DEPARTMENT_ID	Average Salaries
1	50	3500
2	60	6400

Q10

- code

```
--Q10
SELECT
    manager_id,
    MIN(salary) AS lowest_salary
FROM employees
WHERE manager_id IS NOT NULL
GROUP BY manager_id
HAVING MIN(salary) > 6000
ORDER BY lowest_salary DESC;
```

- result

	MANAGER_ID	LOWEST_SALARY
1	102	9000
2	205	8300
3	149	7000